Where is "Washing Machine" Stored in LLMs? Sequential Construction, Not Holistic Storage, Underlies Compound Concept Representation

Ari Holtzman and NeuriCo

Abstract

Large language models must represent millions of compound concepts—"washing machine," "hot dog," "red herring"—yet their residual streams have limited dimensionality. How do these multi-token concepts fit? We investigate whether compound nouns are stored as dedicated directions in the residual stream or are constructed compositionally from their parts. Through three complementary experiments on GPT-2 SMALL (124M parameters), we examine 21 compound concepts spanning the compositionality spectrum from transparent ("kitchen chair") to idiomatic ("red herring"). We find that compound representations are highly similar to their head nouns in the residual stream (mean cosine similarity 0.954), yet sparse autoencoder (SAE) decomposition reveals that 56.3% of features active for compounds are unique to the compound context—invisible to cosine similarity. The degree of unique representation scales with idiomaticity: "red herring" activates 82.1% unique features versus 46.9% for "kitchen chair" (Mann-Whitney $p = 0.0005$, Cohen's $d = 2.54$). Most strikingly, next-token prediction serves as the dominant storage mechanism: after "The washing," the model assigns 47.1% probability to "machine" (a $4{,}221\times$ increase over baseline), effectively constructing the compound concept through sequential prediction rather than holistic representation. These findings reveal that LLMs manage the "too many concepts, too few dimensions" problem by constructing multi-token concepts sequentially, with implications for interpretability, concept editing, and SAE-based analysis.

1 Introduction

How does a language model "know" that a washing machine is a household appliance? GPT-2 Small has 768 dimensions in its residual stream, yet English contains millions of referenceable compound concepts—washing machine, coffee table, red herring, guinea pig. Even the superposition hypothesis [Elhage et al., 2022], which allows models to store $\sim 10\times$ more features than dimensions through near-orthogonal packing, cannot accommodate this many dedicated compound directions. Something must give: either compounds share directions with their components, or the model uses a different mechanism entirely.

We ask: are compound concepts stored as dedicated representations, or are they composed from their parts? Three possibilities exist. First, the model may develop *holistic* representations—unique directions for "washing machine" that are distinct from both "washing" and "machine." Second, the model may rely on *compositional construction*, where seeing "washing" contextually modifies the "machine" representation and makes "machine" overwhelmingly likely as the next token, with no unified compound direction needed. Third, a *hybrid* mechanism may operate, where subtle compound-specific features coexist with largely compositional representations.

Prior work has studied superposition for individual features [Elhage et al., 2022], feature absorption in sparse autoencoders [Chanin et al., 2024], and compound noun semantics in BERT-family models [Ormerod et al., 2024, Miletić and Schulte im Walde, 2023]. However, no study has directly

measured how compound concepts are decomposed by SAEs in autoregressive models, or whether next-token prediction serves as the primary mechanism for compound concept "storage." Park et al. [2024] tested the linear representation hypothesis on 27 single-token concepts but never multi-token compounds. Merullo et al. [2023] showed that LLMs implement vector arithmetic for relational tasks, but did not examine whether compound nouns follow the same compositional pattern.

We address this gap with three complementary experiments on GPT-2 SMALL (124M parameters, 12 layers, 768 dimensions), examining 21 compound concepts that span the full compositionality spectrum. Our experiments measure (1) residual stream cosine similarity between compound and component representations across all layers, (2) SAE feature overlap between compounds and their parts using pre-trained SAEs with 24,576 features per layer, and (3) next-token prediction priming—how strongly a modifier like "washing" shifts the probability distribution toward its compound head "machine."

Our results support the hybrid hypothesis, with composition as the dominant mechanism. The compound direction is nearly identical to the head noun direction in the residual stream (cosine similarity 0.954), yet SAE decomposition reveals that 56.3% of active features are unique to the compound context. The primary mechanism is next-token prediction priming: after "The washing," GPT-2 assigns 47.1% probability to "machine" as the top prediction—a $4{,}221\times$ increase over baseline. Idiomaticity predicts the degree of unique representation: "red herring" activates 82.1% unique SAE features while "kitchen chair" activates only 46.9%.

In summary, we make the following contributions:

- We conduct the first systematic comparison of compound concept representations across three levels of analysis—residual stream geometry, SAE feature decomposition, and next-token prediction—revealing that these levels tell fundamentally different stories about compound storage.
- We show that cosine similarity is a blunt instrument for detecting compound-specific processing: despite 0.954 mean cosine similarity between compounds and head nouns, SAE features reveal 56.3% unique activation, and this unique fraction correlates strongly with linguistic compositionality (Mann-Whitney $p = 0.0005$, Cohen's $d = 2.54$).
- We demonstrate that next-token prediction priming is the dominant mechanism for compound concept construction, with priming ratios spanning five orders of magnitude (from $1.3\times$ for "office desk" to $28{,}037\times$ for "guinea pig"), providing a quantitative taxonomy of how "frozen" different compounds are in the model's predictions.

2 Related Work

Superposition and the dimensionality bottleneck. Elhage et al. [2022] established that neural networks store more features than they have dimensions by packing sparse features as nearly-orthogonal directions—the superposition hypothesis. Scherlis et al. [2022] showed that polysemanticity emerges naturally from capacity constraints. These results imply that compound concepts, being relatively sparse in training data compared to their constituent words, are likely stored in superposition with related concepts. Our work tests a more specific question: whether compound concepts have *any* dedicated direction (even in superposition) or are entirely derived from constituent directions augmented by next-token prediction.

Linear representations in LLMs. The linear representation hypothesis posits that concepts correspond to directions in a model's activation space. Park et al. [2024] formalized three variants and tested 27 single-token concepts in Llama-2, finding strong linear structure for most. Gurnee and Tegmark [2023] showed that LLMs develop linear representations of space and time. Merullo et al. [2023] demonstrated that LLMs implement Word2Vec-style vector arithmetic for relational tasks, suggesting that compositional operations are possible within the residual stream. Unlike these studies, which focus on single-token concepts or binary relations, we directly test whether *multi-token* compound concepts have independent directions or are compositionally derived from their parts.

Sparse autoencoders and feature decomposition. Sparse autoencoders (SAEs) decompose polysemantic activations into more interpretable features [Cunningham et al., 2023, Gao et al., 2024, Rajamanoharan et al., 2024]. However, Chanin et al. [2024] identified the feature absorption problem: when features form hierarchies, SAEs absorb parent features into child features, creating unreliable classifiers. Minegishi et al. [2025] showed that SAEs can separate different word senses

Compound	Modifier	Head	Category	Comp.
washing machine	washing	machine	appliance	medium
hot dog	hot	dog	food	low
red herring	red	herring	idiom	very_low
kitchen chair	kitchen	chair	furniture	very_high
swimming pool	swimming	pool	structure	high
guinea pig	guinea	pig	animal	low
dark horse	dark	horse	idiom	very_low
coffee machine	coffee	machine	appliance	high
school bus	school	bus	vehicle	high
ice cream	ice	cream	food	medium

Table 1: Representative compounds from our dataset with compositionality ratings. Full dataset contains 21 compounds spanning appliances, food, vehicles, idioms, furniture, and clothing categories.

for polysemous words, with sense disambiguation concentrated in deeper layers. Giglemiani et al. [2024] demonstrated that arbitrary combinations of SAE latents do not produce realistic activations, suggesting significant geometric structure beyond simple addition. We use SAE feature analysis not as ground truth but as a complementary lens: the finding that 56% of compound features are unique provides evidence of compound-specific computation that cosine similarity misses, while acknowledging that SAE artifacts may affect exact percentages.

Compound noun semantics in transformers. Ormerod et al. [2024] used representational similarity analysis to compare transformer representations of compound nouns with human judgments in BERT-family models, finding that compounds processed together have different representations than constituents processed separately. Miletić and Schulte im Walde [2023] tested 41,496 BERT configuration combinations for predicting compound compositionality, finding that early layers and surrounding context are most informative. Nedumpozhimana and Kelleher [2024] surveyed multiword expression processing in transformers, confirming that these models encode MWE semantics but with heavy dependence on extraction method and layer choice. Building on these findings, we extend compound analysis to autoregressive models and introduce SAE feature decomposition and next-token priming as new analytical tools, providing a more fine-grained picture than residual stream geometry alone.

Combinatorial interpretability. Adler et al. [2025] proposed feature channel coding as an alternative to SAE-based analysis, where compound features are encoded via overlapping neuron groups in MLP weight matrices. This framework predicts that compound concepts should be represented through intersecting feature channels rather than dedicated directions—consistent with our finding that compounds are contextual modifications of component representations rather than independent directions.

3 Methodology

We conduct three experiments on GPT-2 SMALL [Radford et al., 2019] (124M parameters, 12 layers, 768 dimensions), accessing internal activations via TRANSFORMERLENS [Nanda and Bloom, 2022] and using pre-trained SAEs (24,576 features per layer) from the GPT-2 Small SAE suite. All experiments run on a single NVIDIA RTX A6000 (48GB) in approximately 3 minutes total.

3.1 Dataset Construction

We construct a dataset of 21 compound concepts spanning the compositionality spectrum, with ratings from *very_low* (fully idiomatic, e.g., "red herring") to *very_high* (fully transparent, e.g., "kitchen chair"). table 1 shows representative examples.

Each compound is processed in the minimal context "The [compound]" to isolate the compound representation from richer contextual effects. For component baselines, we process "The [modifier]" and "The [head]" separately.

Tokenization considerations. GPT-2's BPE tokenizer handles these compounds variably: "washing machine" splits cleanly into two tokens (`washing`, `␣machine`), but "coffee machine" becomes three tokens (`co`, `ffee`, `␣machine`) and "red herring" splits the head noun (`red`, `␣her`, `ring`). For multi-token components, the model must already perform compositional processing just to represent the component word, adding complexity to the analysis. We extract representations at the last token position for all conditions.

3.2 Experiment 1: Residual Stream Cosine Similarity

For each compound concept, we measure how the compound representation relates to its components in the residual stream at every layer. Specifically, at each layer ℓ, we compute three cosine similarities:

$$\cos_{\text{head}} = \frac{\boldsymbol{h}^{(\ell)}_{\text{compound}} \cdot \boldsymbol{h}^{(\ell)}_{\text{head}}}{\|\boldsymbol{h}^{(\ell)}_{\text{compound}}\| \|\boldsymbol{h}^{(\ell)}_{\text{head}}\|} \tag{1}$$

$$\cos_{\text{add}} = \frac{\boldsymbol{h}^{(\ell)}_{\text{compound}} \cdot \bar{\boldsymbol{h}}^{(\ell)}_{\text{avg}}}{\|\boldsymbol{h}^{(\ell)}_{\text{compound}}\| \|\bar{\boldsymbol{h}}^{(\ell)}_{\text{avg}}\|} \tag{2}$$

$$\cos_{\text{mod}} = \frac{\boldsymbol{h}^{(\ell)}_{\text{compound}} \cdot \boldsymbol{h}^{(\ell)}_{\text{mod}}}{\|\boldsymbol{h}^{(\ell)}_{\text{compound}}\| \|\boldsymbol{h}^{(\ell)}_{\text{mod}}\|} \tag{3}$$

where $\boldsymbol{h}^{(\ell)}_{\text{compound}}$ is the hidden state at the last token of the compound, $\boldsymbol{h}^{(\ell)}_{\text{head}}$ and $\boldsymbol{h}^{(\ell)}_{\text{mod}}$ are the hidden states for the head noun and modifier processed in isolation, and $\bar{\boldsymbol{h}}^{(\ell)}_{\text{avg}} = \frac{1}{2}(\boldsymbol{h}^{(\ell)}_{\text{mod}} + \boldsymbol{h}^{(\ell)}_{\text{head}})$ is the additive composition baseline.

3.3 Experiment 2: SAE Feature Decomposition

We use pre-trained SAEs at layers 1, 6, and 11 to decompose residual stream activations into sparse feature sets. Each SAE has 24,576 features per layer. For each compound and its components, we:

1. Encode the residual stream activation through the SAE encoder.
2. Identify active features (activation > 0).
3. Compute the fraction of compound features that are *unique to the compound context*—not shared with either the head or modifier processed alone.
4. Compute Jaccard similarity $J = |F_{\text{compound}} \cap F_{\text{head}}| / |F_{\text{compound}} \cup F_{\text{head}}|$ between compound and head feature sets.

This decomposition tests whether the subtle modifications that cosine similarity misses correspond to genuine differences in feature activation patterns.

3.4 Experiment 3: Next-Token Prediction Priming

For each compound "modifier head," we measure how strongly the modifier primes the head noun in the model's next-token predictions:

$$\text{Priming Ratio} = \frac{P(\text{head} \mid \text{“The modifier”})}{P(\text{head} \mid \text{“The”})} \tag{4}$$

where the numerator is the probability of the head noun after seeing the modifier in context, and the denominator is its baseline probability after seeing only "The." We also record the rank of the head noun among all 50,257 vocabulary predictions. This experiment directly tests whether the model "stores" compound concepts primarily through transition probabilities rather than dedicated feature directions.

4 Results

4.1 Residual Stream Similarity: Compounds Are Modifications, Not New Directions

table 2 presents the cosine similarity between compound and component representations at the final layer (layer 11). The mean similarity between compound and head noun representations is

Compound	Comp. Level	vs. Head Alone	vs. Additive	vs. Modifier
school bus	high	**0.983**	0.966	0.823
swimming pool	high	0.961	0.965	0.810
coffee machine	high	0.957	0.967	0.844
kitchen chair	very_high	0.956	0.968	0.831
washing machine	medium	0.951	0.954	0.812
guinea pig	low	0.942	0.938	0.760
ice cream	medium	0.937	0.943	0.762
hot dog	low	0.936	0.943	0.761
dark horse	very_low	0.936	0.947	0.777
red herring	very_low	0.895	0.925	0.748
Mean (all 21)	—	0.954 ± 0.019	0.959 ± 0.013	—

Table 2: Cosine similarity between compound representations and component representations at layer 11. All compounds maintain high similarity (> 0.89) with their head nouns, indicating that compound representations are modifications of existing component directions rather than new directions. Best per-row result in **bold** when noteworthy.

Compound	Comp.	Unique (%)	Head Overlap (%)	Mod. Overlap (%)	Jaccard w/ Head
red herring	very_low	**82.1**	14.3	7.1	0.079
dark horse	very_low	78.3	19.3	13.3	0.147
hot dog	low	66.7	23.5	15.7	0.152
guinea pig	low	61.7	31.7	18.3	0.209
washing machine	medium	57.9	35.1	24.6	0.256
coffee machine	high	54.7	34.4	20.3	0.265
swimming pool	high	53.7	33.3	20.4	0.231
school bus	high	48.9	46.7	28.9	0.333
kitchen chair	very_high	46.9	34.7	32.7	0.205
dish washer	high	21.5	**69.2**	27.7	**0.479**
Mean (all 21)	—	56.3 ± 13.1	—	—	0.255 ± 0.091

Table 3: SAE feature decomposition at layer 11. "Unique" is the fraction of compound features not shared with either component alone. "Head/Mod. Overlap" is the fraction of compound features also active for the head/modifier in isolation. Idiomatic compounds activate substantially more unique features than transparent ones. Best (most extreme) values in **bold**.

0.954 ± 0.019, and between compound and additive composition is 0.959 ± 0.013. Even the least compositional compound in our dataset, "red herring," maintains a cosine similarity of 0.895 with its head noun alone.

Compositionality predicts residual stream similarity. There is a significant positive correlation between compositionality and similarity with the additive composition baseline (Spearman $\rho = 0.602$, $p = 0.004$): transparent compounds like "kitchen chair" (0.968) are closer to the additive baseline than idiomatic compounds like "red herring" (0.925). However, even the most idiomatic compounds show very high cosine similarity, suggesting that residual stream geometry provides a limited window into compound-specific processing.

4.2 SAE Features Reveal Hidden Compound-Specific Computation

Despite high cosine similarity at the residual stream level, SAE decomposition reveals that a majority of features active for compounds are not shared with either component alone. table 3 presents the SAE feature analysis at layer 11. The mean fraction of unique-to-compound features is 0.563 ± 0.131, and the mean Jaccard similarity between compound and head feature sets is only 0.255 ± 0.091.

Compositionality strongly predicts unique feature fraction. Low-compositionality compounds (very_low and low) have significantly more unique SAE features (mean 71.2%) than high-compositionality compounds (mean 48.2%). A Mann-Whitney U test confirms this difference

Compound	$P(\text{head} \mid \text{mod})$	$P(\text{head} \mid \text{baseline})$	Priming Ratio	Rank
guinea pig	0.6624	0.000024	**28,037×**	**1**
vending machine	0.5416	0.000112	4,853×	**1**
washing machine	0.4711	0.000112	4,221×	**1**
swimming pool	0.3184	0.000079	4,022×	**1**
sewing machine	0.3741	0.000112	3,352×	**1**
ice cream	0.0743	0.000020	3,757×	3
slot machine	0.0624	0.000112	559×	2
hot dog	0.0307	0.000127	242×	5
dark horse	0.0097	0.000067	144×	8
kitchen chair	0.0012	0.000064	19×	83
red herring	0.0019	0.000434	4.3×	67
rain coat	0.0001	0.000015	3.9×	888
office desk	0.0000	0.000017	1.3×	1429

Table 4: Next-token prediction priming results. The priming ratio (equation 4) measures how much more likely the head noun is after the modifier compared to baseline. "Frozen" compounds like "guinea pig" and "washing machine" show extreme priming ($> 4{,}000\times$), while loosely associated compounds like "office desk" show minimal priming ($1.3\times$). Rank indicates the head noun's position among all 50,257 vocabulary predictions.

is statistically significant ($p = 0.0005$), with a very large effect size (Cohen's $d = 2.54$). This makes linguistic sense: idioms like "red herring" require more specialized representation because their meaning cannot be derived from their components.

"Dish washer" is an informative outlier. With only 21.5% unique features and 69.2% overlap with "washer" alone, "dish washer" is the most compositional compound by SAE metrics—the model treats it almost identically to "washer." This is semantically reasonable: a dishwasher *is* a type of washer, so minimal additional representation is needed.

Cosine similarity versus SAE decomposition. The contrast between these two analysis levels is striking. "Red herring" has 0.895 cosine similarity with "herring" alone (seemingly very similar), yet only 14.3% feature overlap. This demonstrates that cosine similarity measures overall direction alignment while SAE decomposition captures fine-grained feature-level differences invisible to coarser geometric measures.

4.3 Next-Token Priming: The Primary Storage Mechanism

table 4 presents the next-token prediction priming results. The variation across compounds is enormous, spanning five orders of magnitude from $1.3\times$ ("office desk") to $28{,}037\times$ ("guinea pig").

Three regimes of compound priming. The data naturally cluster into three groups:

1. *Extreme priming* ($> 1{,}000\times$): washing machine, vending machine, sewing machine, swimming pool, guinea pig. These are "frozen" compounds where the modifier almost exclusively precedes its head in natural text. The head noun is the #1 prediction.
2. *Moderate priming* (10–$1{,}000\times$): hot dog, slot machine, dark horse, ice cream. The modifier has other common continuations, but the compound is still strongly predicted.
3. *Minimal priming* ($< 10\times$): red herring ($4.3\times$), rain coat ($3.9\times$), office desk ($1.3\times$). These modifiers have many common continuations and the compound is either rare or not strongly associated.

"Washing machine" is constructed through prediction. After "The washing," GPT-2 produces the following top-5 predictions: `machine` (47.1%), `of` (8.4%), `-` (8.2%), `machines` (4.9%), `up` (3.8%). The model assigns nearly half its probability mass to "machine," effectively constructing the compound concept through its transition probabilities. This is the primary mechanism by which GPT-2 "stores" the concept of a washing machine: not as a dedicated direction, but as an overwhelming statistical association.

"Guinea pig" is the most frozen compound. Despite being rated as low compositionality (the meaning of "guinea pig" is opaque), "guinea" predicts "pig" with 66.2% probability and a $28{,}037\times$ priming ratio. This dissociation between semantic compositionality and statistical predictability reveals that priming reflects co-occurrence frequency, not semantic transparency.

5 Discussion

5.1 Three Levels Tell Three Stories

Our three experiments converge on a consistent but nuanced picture: compound concepts in GPT-2 SMALL are *not* stored as dedicated directions, but they are also not *merely* compositional. The three analysis levels reveal complementary aspects of compound representation:

1. *Residual stream geometry* (cosine similarity 0.954): The compound representation is a subtle modification of the head noun direction. From the perspective of overall direction, "washing machine" looks like "machine."
2. *SAE feature decomposition* (56.3% unique features): This subtle modification activates a largely different set of sparse features. The compound modifies the representation in ways that are invisible to cosine similarity but detectable by SAE decomposition.
3. *Next-token prediction* ($4{,}221\times$ priming): The model's primary mechanism for compound storage is sequential prediction. The concept of "washing machine" is encoded not as a direction but as a transition probability.

This three-level picture resolves the apparent paradox: how can the compound be "almost the same" as the head noun (0.954 cosine similarity) yet activate mostly different SAE features (56.3% unique)? The answer is that cosine similarity measures the overall direction of a 768-dimensional vector, while SAE features capture fine-grained structure along specific feature directions. A small rotation in the overall direction can correspond to large changes in which features are active, because SAE features operate near their activation thresholds.

5.2 The Role of Compositionality

The correlation between linguistic compositionality and unique SAE features (Cohen's $d = 2.54$) reveals that the model allocates more specialized representation to semantically opaque compounds. "Red herring" activates 82.1% unique features because the idiomatic meaning (a misleading clue) cannot be derived from "red" and "herring" separately—the model must encode this non-compositional semantics through additional features. In contrast, "kitchen chair" (46.9% unique) is nearly derivable from its parts, requiring less specialized representation.

An intriguing pattern emerges from the priming data: "guinea pig" has the highest priming ratio ($28{,}037\times$) but moderate unique features (61.7%), while "red herring" has the most unique features (82.1%) but minimal priming ($4.3\times$). This dissociation reflects two independent dimensions of compound representation: statistical predictability (how frozen the compound is in text) and semantic opacity (how much the compound meaning deviates from component meanings). "Guinea pig" is statistically frozen but semantically processed compositionally by the SAE, while "red herring" is statistically free but semantically specialized.

5.3 Implications

For mechanistic interpretability. Our findings challenge the assumption that cosine similarity is sufficient for detecting concept-specific processing. Two representations with 0.895 cosine similarity ("red herring" vs. "herring") can have only 14.3% feature overlap, suggesting that interpretability methods relying solely on geometric measures may miss important structure. SAE decomposition provides a complementary and more sensitive lens, though one must account for potential SAE artifacts [Chanin et al., 2024].

For the superposition hypothesis. Compound concepts illustrate how LLMs manage the "too many concepts, too few dimensions" problem. Rather than allocating dedicated directions for each of the millions of possible compounds, the model constructs multi-token concepts sequentially through transition probabilities. This is a fundamentally different strategy from superposition: instead of

packing more features into the same space, the model distributes concept representation across *time steps*. Superposition handles the single-token feature bottleneck; sequential construction handles the multi-token concept bottleneck.

For concept editing. Editing compound concept knowledge (e.g., removing the model's knowledge that a washing machine is an appliance) would likely require intervention at multiple levels: modifying the transition probabilities (attention patterns and FFN weights that produce the $4{,}221\times$ priming effect) *and* the compound-specific SAE features (the 57.9% unique features for "washing machine"). Modifying a single direction is unlikely to suffice, given that compound representations are distributed across sequential prediction and fine-grained feature modulation.

5.4 Limitations

Single model. We study GPT-2 SMALL (124M parameters) only. Larger models with more dimensions may allocate dedicated directions for common compounds. The absence of holistic compound representations in GPT-2 may reflect capacity constraints rather than a general principle.

Minimal context. We use the context "The [compound]" for all experiments. Richer contexts (e.g., "She loaded the washing machine with clothes") could produce different activation patterns. Natural corpus contexts from large-scale text would better capture the range of compound usage.

SAE artifacts. Pre-trained SAEs have known issues including feature absorption [Chanin et al., 2024] and polysemantic features. The 56.3% unique feature fraction might partly reflect SAE reconstruction artifacts rather than genuine compound-specific computation. However, the strong correlation between unique features and compositionality ($d = 2.54$) suggests that the pattern is meaningful even if exact percentages are noisy.

Hand-assigned compositionality ratings. Our compositionality labels are hand-assigned rather than derived from human judgment experiments. Validated compositionality ratings from psycholinguistic studies [Reddy et al., 2011] would strengthen the compositionality analysis.

Tokenization effects. Some compounds split unevenly under BPE tokenization (e.g., "coffee" $\rightarrow$ `co` + `ffee`), meaning the "modifier" representation already involves compositional processing. This complicates the interpretation of feature overlap statistics for multi-token modifiers.

6 Conclusion

We investigated how compound concepts like "washing machine" are represented in the residual stream of GPT-2 SMALL, examining 21 compounds across the compositionality spectrum. Our three experiments reveal that compound concepts are not stored as dedicated directions. Instead, the model uses three complementary mechanisms: (1) contextual modification of head noun representations (cosine similarity 0.954), (2) compound-specific SAE features that capture fine-grained differences invisible to geometric measures (56.3% unique features), and (3) massive next-token prediction priming that constructs compounds sequentially ($4{,}221\times$ for "washing machine").

The key takeaway is that LLMs solve the "too many concepts, too few dimensions" bottleneck for compound concepts by distributing representation across time steps rather than packing more features into the same vector space. This is a distinct strategy from superposition: where superposition handles single-token features, sequential construction handles multi-token concepts.

Future work should extend this analysis to larger models where dedicated compound directions may emerge, test with richer natural contexts, apply causal interventions to establish whether compound-specific SAE features are functionally necessary, and explore whether the sequential construction pattern holds across languages with different compounding productivity.

References

Sun Adler, Aditya Bhaskar, Surya Ganesh, and Lijie Chen. Towards combinatorial interpretability. *arXiv preprint arXiv:2504.08842*, 2025.

David Chanin, Fazl Barez, Adam Karvonen, and Neel Nanda. A is for absorption: Studying feature splitting and absorption in sparse autoencoders. *arXiv preprint arXiv:2409.14507*, 2024.

Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models. *arXiv preprint arXiv:2309.08600*, 2023.

Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, et al. Toy models of superposition. *arXiv preprint arXiv:2209.10652*, 2022.

Leo Gao, Tom Dupré la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. Scaling and evaluating sparse autoencoders. *arXiv preprint arXiv:2406.04093*, 2024.

Giorgi Giglemiani, Samuel Marks, and Neel Nanda. Evaluating synthetic activations composed of SAE latents in GPT-2. *arXiv preprint arXiv:2409.15019*, 2024.

Wes Gurnee and Max Tegmark. Language models represent space and time. *arXiv preprint arXiv:2310.02207*, 2023.

Jack Merullo, Carsten Eickhoff, and Ellie Pavlick. Language models implement simple word2vec-style vector arithmetic. *arXiv preprint arXiv:2305.16130*, 2023.

Filip Miletić and Sabine Schulte im Walde. A systematic search for compound semantics in pretrained BERT architectures. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, 2023.

Kei Minegishi, Hiroshi Noji, and Yohei Oseki. Rethinking evaluation of sparse autoencoders through the representation of polysemous words. *arXiv preprint arXiv:2501.06254*, 2025.

Neel Nanda and Joseph Bloom. TransformerLens. 2022. `https://github.com/neelnanda-io/TransformerLens`.

Vasudevan Nedumpozhimana and John D. Kelleher. Semantics of multiword expressions in transformer-based models: A survey. *arXiv preprint arXiv:2407.16751*, 2024.

Mark Ormerod, Jesús Maldonado, and Roman Klinger. How is a "kitchen chair" like a "farm horse"? exploring the representation of compound nominals in transformer-based language models. *Computational Linguistics*, 50(1), 2024.

Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the geometry of large language models. *arXiv preprint arXiv:2311.03658*, 2024.

Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. Language models are unsupervised multitask learners. *OpenAI blog*, 2019.

Senthooran Rajamanoharan, Arthur Conmy, Lewis Smith, Tom Lieberum, Vikrant Varma, János Kramár, Rohin Shah, and Neel Nanda. Improving dictionary learning with gated sparse autoencoders. *arXiv preprint arXiv:2404.16014*, 2024.

Siva Reddy, Diana McCarthy, and Suresh Manandhar. An empirical study on compositionality in compound nouns. *Proceedings of IJCNLP*, 2011.

Adam Scherlis, Kshitij Sachan, Adam S Jermyn, Joe Benton, and Buck Shlegeris. Polysemanticity and capacity in neural networks. *arXiv preprint arXiv:2210.01892*, 2022.